

Small worlds. Deep time.

A walkable natural-history museum.

A complete architectural shell.

A place for your Blender exhibits.

76 × 72

METRE FOOTPRINT

11

CONNECTED SPACES

30

EXHIBIT ORIGINS

Architectural edition / no specimen models

Dinosaur plinths, insect cabinets, empty habitats, a roof-lit connecting court, visitor controls, measured plans and a Blender handoff. The building is original; Melbourne Museum provides the research reference, not an exact floor plan.

Research reviewed 11 September 2026.

Independent project; not affiliated with Museums Victoria.

Virtual-world concept, not a construction or accessibility certification.

Research into spatial decisions

What the source establishes; what ThreeMuseum deliberately changes.

S1 / Bugs Alive!

Reference: The official gallery combines pinned collections, living enclosures, enlarged anatomy and a theatre.

Design response: Separate the bug wing into diversity, habitats, anatomy and arachnids. Use hollow cabinets, not rooms full of generic cubes.

S2-S3 / Dinosaur Walk

Reference: The museum describes visitors moving around and beneath mounted skeletons, including a long-necked Mamenchisaurus.

Design response: Build a tall hall, a split-support centrepiece bay, comparison tables and a raised long-view position. Do not imitate skeletons with placeholder geometry.

S4 / Triceratops

Reference: Melbourne has a separate immersive Triceratops exhibition.

Design response: Create Last Giants as a distinct, quieter feature room with a walk-around mount allowance. Do not reproduce Horridus.

S5 / Official visitor map

Reference: The Science and Life galleries connect with the museum's broader circulation and Forest Gallery.

Design response: Group the two subject areas beside a legible connecting court. The footprint and room dimensions in this atlas are invented design parameters.

S6 / Denton Corker Marshall

Reference: The architect describes a campus of elements rather than a single monumental object.

Design response: Use distinct ceiling heights and volumes: one tall hall, a bright circulation spine and smaller galleries.

S7 / Accessibility

Reference: The museum documents sensory supports and access planning for visitors.

Design response: Provide no head bob or compulsory sound, a lower eye-height option, keyboard look, room-map shortcuts and a quiet theatre.

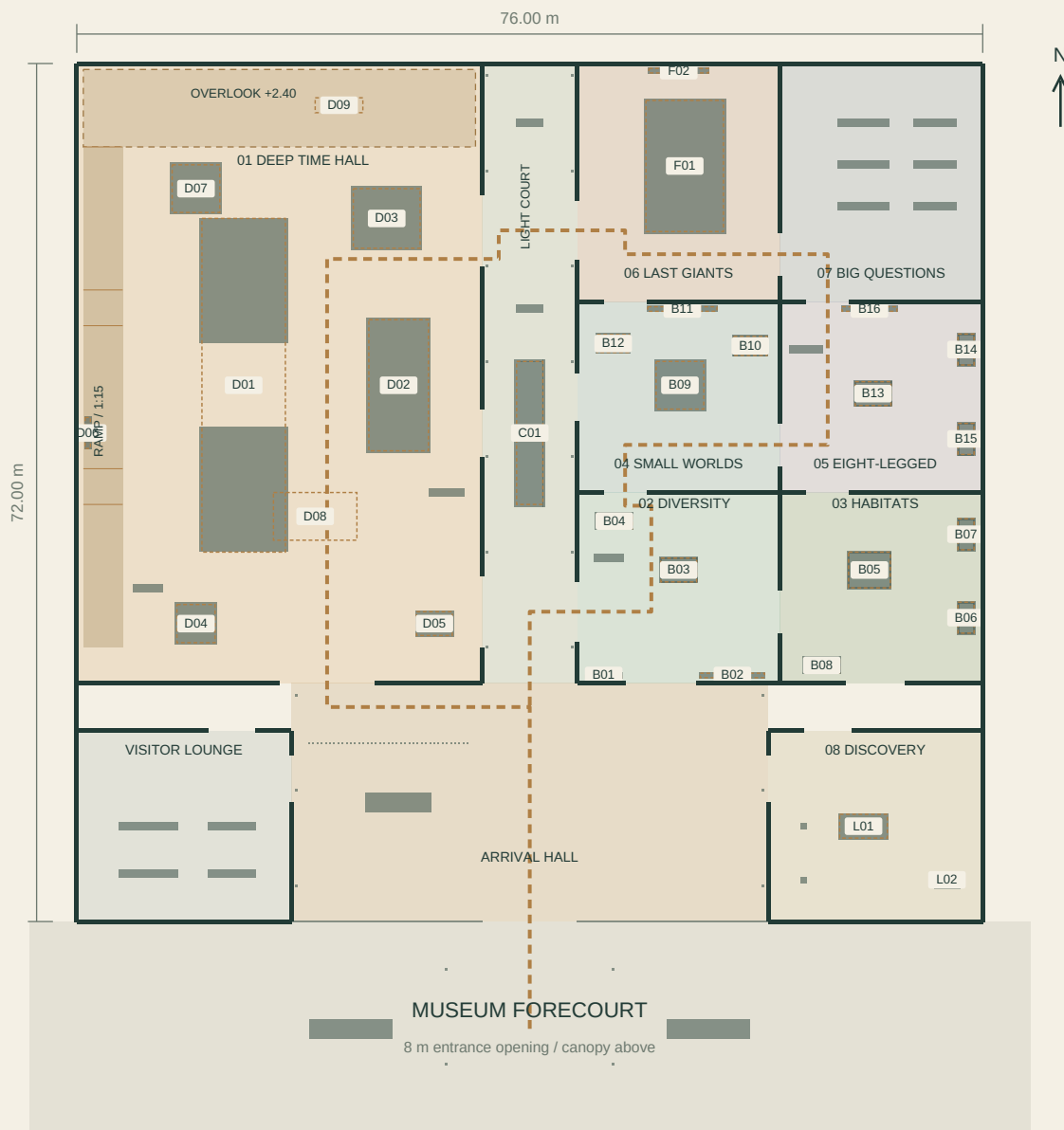
The whole museum

P01 / Ground floor, raised overlook and optional visitor loop.

THREE MUSEUM

P01 / GROUND FLOOR + OVERLOOK

Original design. All dimensions in metres. Not an as-built Melbourne Museum plan.



--- Suggested visitor loop

--- Future model envelope

--- Raised circulation

D = dinosaurs B = bugs F = feature L = learning C = court

0 5 10 m

Use written dimensions; viewer/print size changes scale.

The dashed route was checked against the collision geometry. It is optional: visitors may choose a short bug-only or dinosaur-only visit and use the map to change galleries.

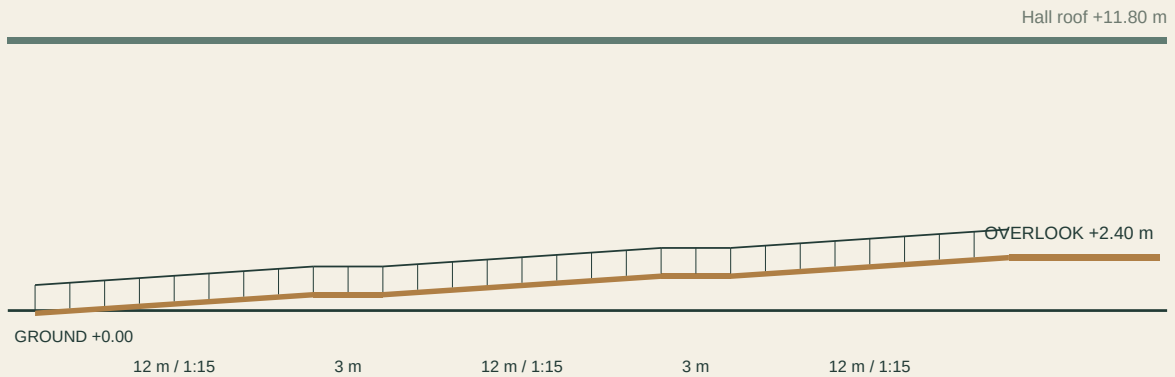
Height, daylight and the long view

P02-P03 / A varied section without stairs on the main route.

THE LONG VIEW

P03 / WEST RAMP LONGITUDINAL SECTION

Same horizontal and vertical scale. Railings shown schematically.



Three 12 m slopes + two 3 m landings = 42 m run. Ramp width at rail centrelines: 3.30 m.

Virtual-world dimensions only. No structural, fire, disability-access or building-code certification is implied.

Room-specific atmosphere

Deep Time Hall: 11.8 m roof datum.

Last Giants: 8.0 m.

Bug rooms: 5.6 m.

Light Court: 10.0 m.

Arrival hall: 9.0 m.

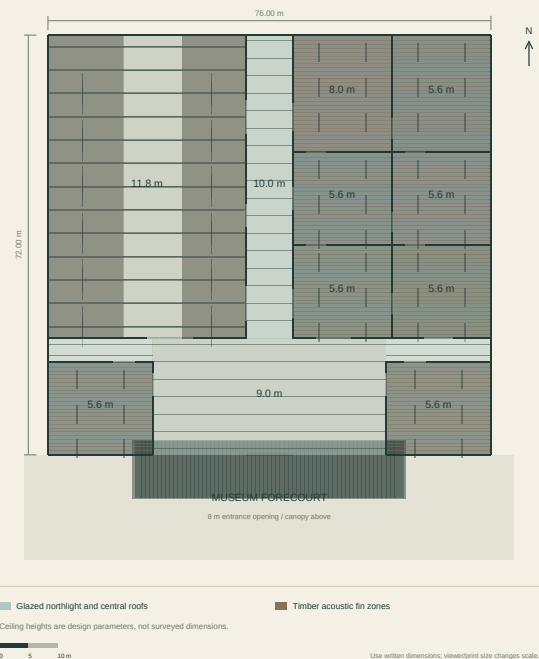
Three 12 m sloping runs rise 0.8 m each. Two 3 m landings interrupt the ascent. The ramp is 3.3 m wide; the overlook is at +2.4 m, with a 1.1 m guardrail datum.

These are game-space design choices. No claim is made that the ramp, glazing, guardrails or exits satisfy a particular building regulation.

THREE MUSEUM

P02 / REFLECTED CEILING CONCEPT

Original design. All dimensions in metres. Not an as-built Melbourne Museum plan.



A building organised for visiting

Room footprints include partition allowances; they are not net lettable areas.

SPACE	DESIGN FOOTPRINT	GROSS AREA	ROOF
Arrival hall	40 × 20 m	800 m ²	9 m
Deep Time Hall	34 × 52 m	1,768 m ²	11.8 m
Light Court	8 × 52 m	416 m ²	10 m
Cabinet of Diversity	17 × 16 m	272 m ²	5.6 m
Living Habitats	17 × 16 m	272 m ²	5.6 m
Small Worlds Lab	17 × 16 m	272 m ²	5.6 m
The Eight-legged Room	17 × 16 m	272 m ²	5.6 m
Last Giants	17 × 20 m	340 m ²	8 m
Big Questions Theatre	17 × 20 m	340 m ²	5.6 m
Discovery Studio	18 × 16 m	288 m ²	5.6 m
Visitor lounge	18 × 16 m	288 m ²	5.6 m

Three useful routes

Short visit: Arrival → Diversity → Living Habitats → Gallery Street → Arrival. **Deep time visit:** Arrival → Deep Time Hall → ramp → overlook → return ramp. **Long loop:** Hall → Light Court → Last Giants → Theatre → arachnid and anatomy rooms → Diversity → Arrival.

The 76 × 72 m footprint is 5,472 m². The room rectangles account for 5,328 m²; the remaining 144 m² is the side portion of Gallery Street. The overlook and ramp are additional raised circulation within the footprint. These figures are derived from the model, not from Melbourne Museum.

Deep Time Hall

Nine mounting and interpretation positions. All are intentionally empty.

The centrepiece is D01, a 28 m-long design allowance with two separate support islands. Its 7 m cross-passage remains physically open. A final model must keep that passage clear above 2.8 m and must not allow its neck, tail or supports to intrude into the adjacent walk. The 28 m envelope is not a claimed specimen measurement.

ID / SUBJECT	FIXTURE	W × H × D (m)
D01 / Mamenchisaurus	split	7 × 8.5 × 28
D02 / Tarbosaurus	plinth	5 × 5.8 × 11
D03 / Armoured dinosaur	plinth	5.6 × 3.5 × 5
D04 / Protoceratops	plinth	3.2 × 2.5 × 3.2
D05 / Trackway casting	table	3 × 0.3 × 2
D06 / Comparative teeth	wall	0.45 × 1.1 × 2.6
D07 / Australian megafauna	plinth	4 × 3.6 × 4
D08 / Pterosaur	suspended	7 × 2.8 × 4
D09 / Fossil comparison	table	4 × 0.45 × 1.25

D08 is suspended; its origin is 7 m above ground. D09 sits on the overlook, with its specimen origin at +3.28 m. Distinguish pterosaurs and later megafauna from dinosaurs in the final interpretation. Scientific labels, species-specific claims and mount authenticity still require review. [S2-S3]

The bug galleries

Sixteen future exhibits, organised by the questions visitors ask.

ID / SUBJECT	FIXTURE	W × H × D (m)
B01 / Butterfly diversity	wall	3 × 1.8 × 0.38
B02 / Beetle diversity	wall	5.4 × 1.8 × 0.38
B03 / Insect Hall of Fame theme	case	3 × 1.65 × 2
B04 / Insect classification	table	3 × 0.4 × 1.25
B05 / Green tree ant colony	habitat	3.5 × 2.4 × 3
B06 / Mantid habitat	habitat	1.35 × 1.8 × 2.6
B07 / Aquatic invertebrates	habitat	1.35 × 1.8 × 2.6
B08 / Decomposer habitat	habitat	3 × 1.8 × 1.25
B09 / Insect body systems	plinth	4 × 2.9 × 4
B10 / Insect mouthparts	case	2.8 × 1.1 × 1.6
B11 / Life cycles	wall	5.8 × 1.8 × 0.38
B12 / Pollination	table	2.7 × 0.7 × 1.5
B13 / Arachnid comparison	case	3 × 1.35 × 2
B14 / Tarantula enclosure	habitat	1.35 × 1.8 × 2.6
B15 / Local spider enclosure	habitat	1.35 × 1.8 × 2.6
B16 / Web structures	wall	4.6 × 1.8 × 0.38

The themes draw on Bugs Alive!, but the rooms and fixture dimensions are new. Habitat cases contain no animals, branches, water or substrate dressing. The tarantula case is a presentation allowance, not a quarantine facility. Microscope, mouthpart and lifecycle models remain a Blender task. [S1]

The supporting spaces

Feature gallery, learning studio, light court and restrained materials.

ID / SUBJECT	FIXTURE	W × H × D (m)
F01 / Triceratops	plinth	6.5 × 5 × 11
F02 / Fossil preparation	wall	5 × 1.8 × 0.38
L01 / Handling collection	table	4 × 0.55 × 2
L02 / Microscope station	table	2 × 0.8 × 1.3
C01 / Botanical installation	planter	2.2 × 6 × 12

Material and lighting direction

Limestone and terrazzo: Quiet, low-gloss bases and circulation floors. Keep fine detail in the texture rather than oversized bump patterns.

Oak and dark bronze: Warmer dinosaur flooring, acoustic ceiling fins, handrails, light tracks and cabinet framing.

Glazing: A lightweight tinted-glass approximation for the baseline web scene. No ray-traced or physically accurate transmission claim.

Daylight and gallery light: Roof-lit arrival and court, a long dinosaur northlight, warm case lamps and a fixed pool of four local fill lights.

Lighting restraint: No heavy fog, bloom, flashing projections or artificial lens dirt. Screen and planting infrastructure are present, but films and plants are absent.

Blender to the exhibit bay

One coordinate system, explicit origins and no automatic placement guesswork.

01 / Build the reference

Run `blender/build_shell.py` in Blender. It reconstructs the architectural geometry from `public/architecture.json`, creates named origins and wire envelopes, and leaves the roof hidden. Save the resulting scene as your own `.blend` file.

02 / Author the content

Put static world-space specimen geometry in `MODEL_D01`, `MODEL_B05`, and so on. Leave the building, display case, labels and wire envelope out of the exhibit collection. All dimensions are metres.

03 / Export

For static models in the reference scene, set `EXHIBIT_ID` in `export_static_exhibit.py` and run it. For animated assets, author/export a self-contained GLB in local coordinates with a bottom-centred origin; see `BLENDER_HANDOFF.md`.

04 / Preview and validate

Press B in the museum, choose a bay, and open the GLB. Set offset, rotation and uniform scale. The fit button is a convenience: it changes presentation size and does not certify scientific scale.

05 / Persist the installation

Copy the GLB to `public/models`, export the manifest from the installation panel and replace `public/exhibits.json`. Browser previews cannot silently save files into the project directory.

Coordinate contract

Three.js: X east, Y up, north is -Z. **Blender:** (x, y, z) = (Three X, -Three Z, Three Y). A slot position is the bottom centre of the exhibit. A GLB scale of 1 means one metre stays one metre. Offsets use metres; UI rotations use degrees.

Imported GLBs do not create physics colliders. Keep them inside their display envelopes and add explicit solids to `architecture.js` when an exhibit truly needs additional collision. Standard GLBs with embedded textures are supported; Draco/KTX2 decoder wiring is not included.

What this build does - and does not

An architectural starting point with explicit validation boundaries.

Implemented

Procedural museum geometry; 30 empty display allowances; roof-off inspection; first-person walking and touch input; wall/case collision with displacement substeps; continuous ramp support; map destinations; readable exhibit briefs; GLB preview and manifest export; editable source; static-server and Pages deployment scripts.

Checked in this delivery environment

22 dependency-free geometry and navigation tests pass. They cover room spawns, front doors, the full drawn visitor loop, display collisions, high-displacement wall blocking, ramp ascent/descent, deck guardrails and safe reading positions. JavaScript and Blender-script syntax checks are separate from runtime rendering.

Not verified here

The environment could not retrieve the Three.js runtime, so live WebGL rendering, shader compilation, visual lighting quality and in-browser GLB installation were not exercised. Blender was not available, so the reference/export scripts were syntax-checked but not executed inside Blender. These are real remaining checks, not claimed passes.

Deliberately outside scope

No dinosaur, insect, arachnid, fossil, plant, microscope or habitat-dressing models. No crowds, film content, soundscape, complete scientific labels, collection-management backend, real animal housing, construction documentation or claim of legal accessibility compliance. No exact replica of Melbourne Museum or its proprietary specimens.

Local acceptance pass

Install dependencies, start the server and check every room at visitor eye height. Walk the ramp both ways, inspect the roof-off plan, preview a small test GLB, reload its exported manifest, and test touch controls on the intended device. Recheck every circulation route after adding large models. Use docs/TEST_REPORT.md as the starting checklist.

Primary research sources

Reviewed 11 September 2026. References inform themes, not surveyed dimensions.

[S1] Museums Victoria

Museums Victoria — Bugs Alive!

<https://museums victoria .com.au/melbournemuseum/whats-on/bugs-alive/>

[S2] Museums Victoria

Museums Victoria — Dinosaur Walk

<https://museums victoria .com.au/melbournemuseum/whats-on/dinosaur-walk/>

[S3] Museums Victoria

Museums Victoria — Dinosaur Walk learning resources

<https://museums victoria .com.au/melbournemuseum/resources/dinosaur-walk/>

[S4] Museums Victoria

Museums Victoria — Triceratops: Fate of the Dinosaurs

<https://museums victoria .com.au/melbournemuseum/whats-on/triceratops-fate-of-the-dinosaurs/>

[S5] Museums Victoria

Museums Victoria — Visitor map (27 July 2026)

<https://museums victoria .com.au/media/ux5heeeh/melbourne-museum-map-27072026.pdf>

[S6] Denton Corker Marshall

Denton Corker Marshall — Melbourne Museum

<https://dentoncorkermarshall.com/projects/melbourne-museum/>

[S7] Museums Victoria

Museums Victoria — Accessibility

<https://museums victoria .com.au/melbournemuseum/plan-your-visit/accessibility/>

[S8] Three.js

Three.js — GLTFLoader

<https://threejs.org/docs/pages/GLTFLoader.html>

The official visitor map was inspected as a spatial reference. Source imagery, source maps, exhibit text and specimen assets are not redistributed. All plans in this booklet are generated from ThreeMuseum's own architecture records.